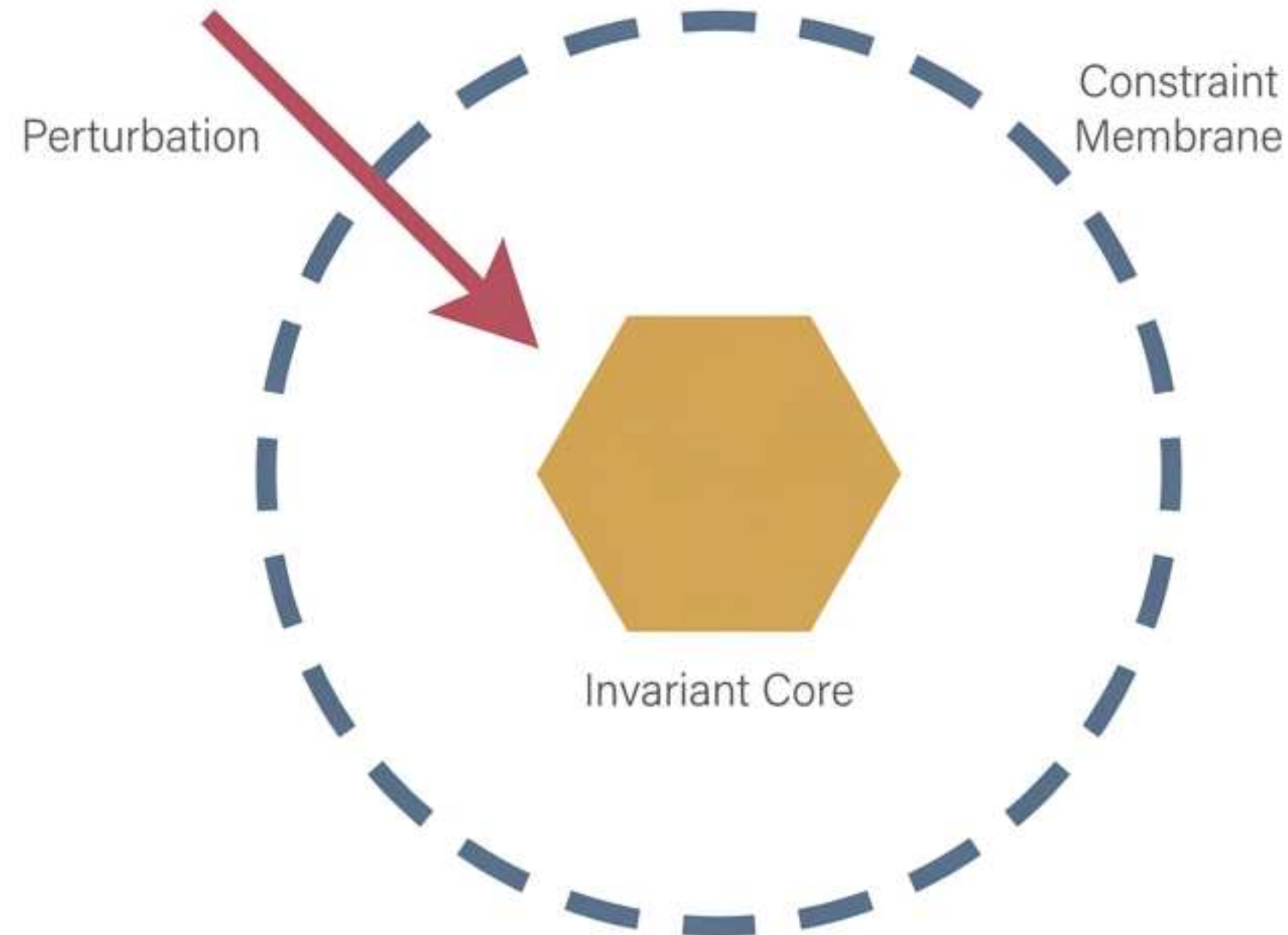


The Admissibility Principle

Computation, Physics, and Analogy as One Operation

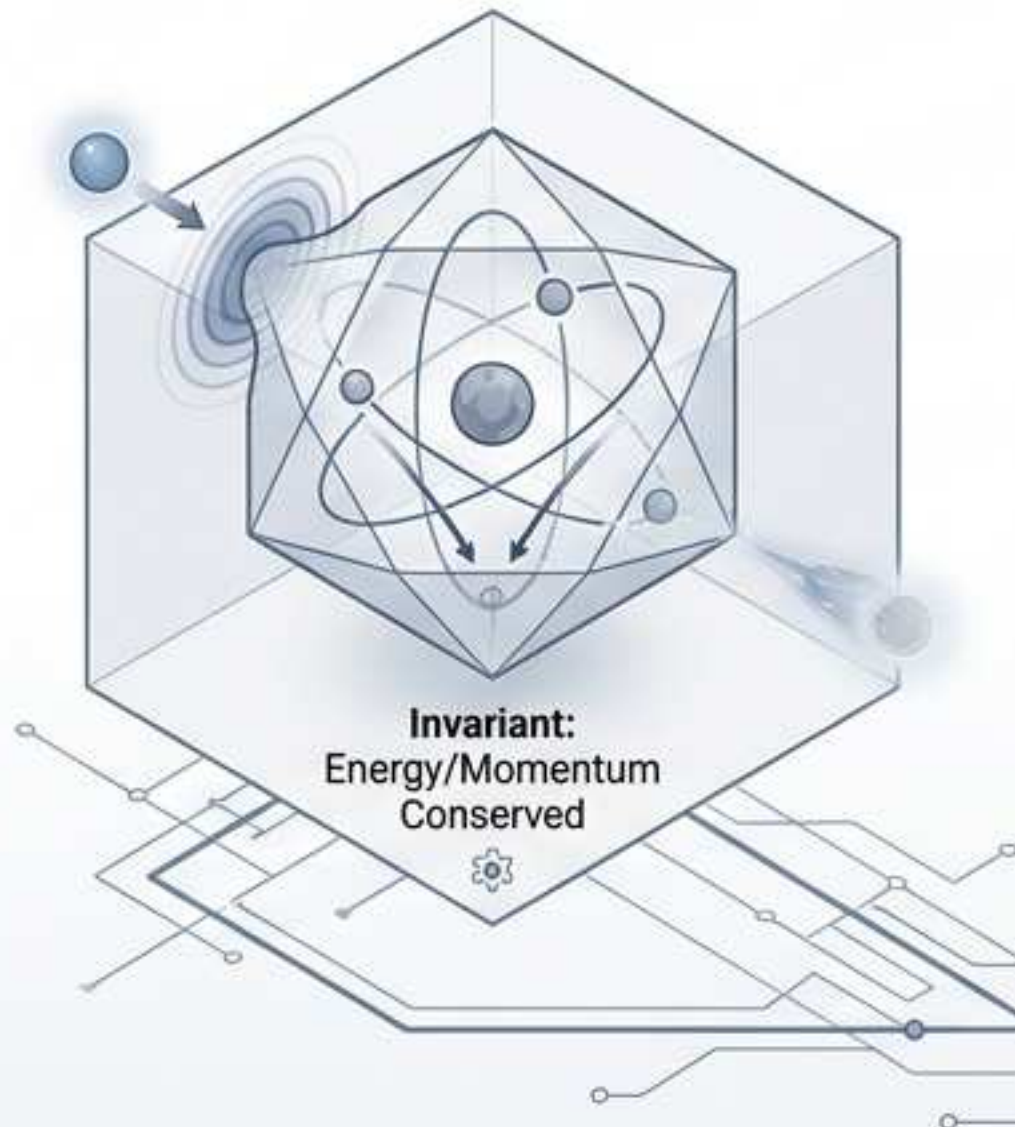


A Shared Formal Skeleton Across Distant Sciences

Standard syntheses fail via **premature literalism** or **vacuous analogy**. The Admissibility Principle claims these domains share a single **formal grammar**: a bounded region tests a perturbation, reorganizes its constraints, and collapses into a new state while preserving an invariant.

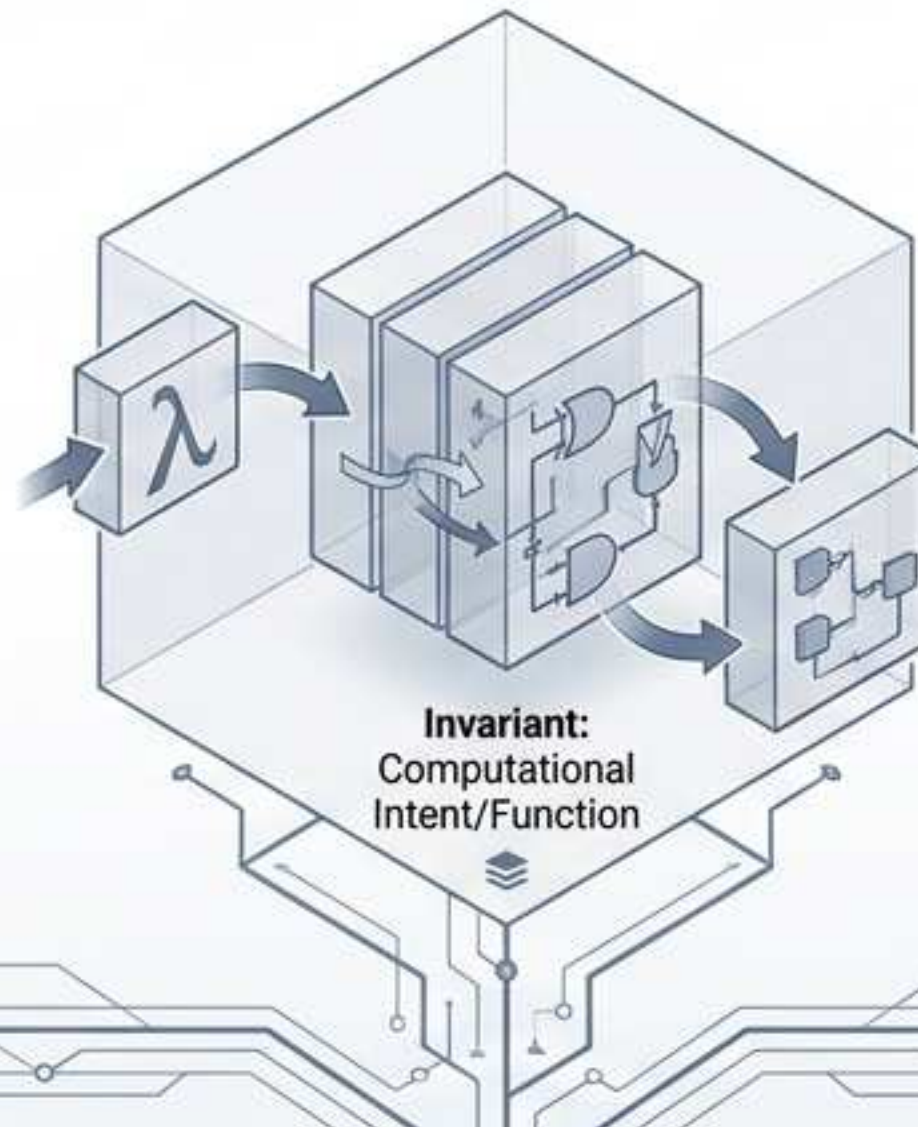
Quantum Physics

What does a photon do when it briefly becomes an atomic excitation?



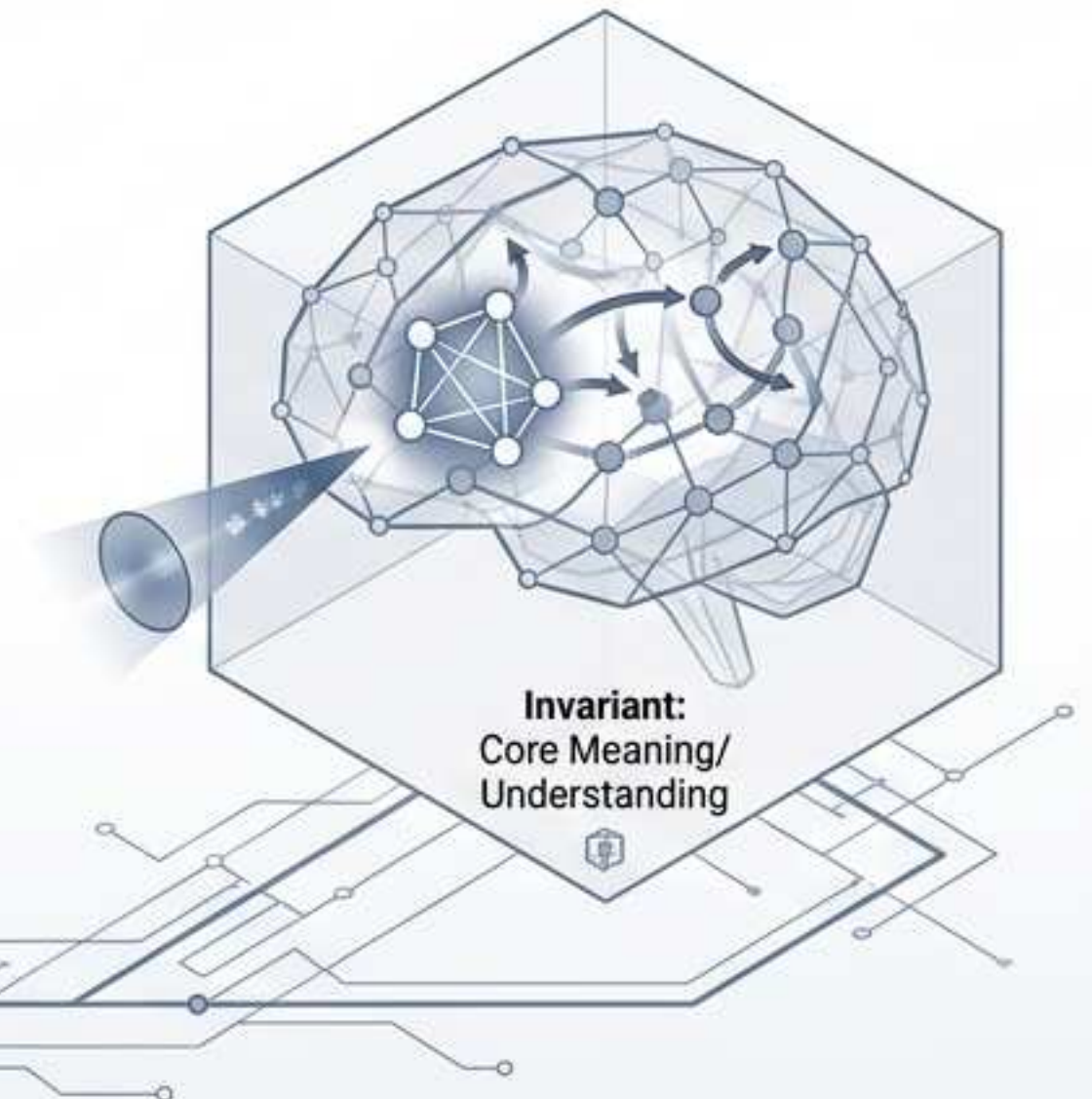
Computation

What does a λ -term do when it swallows its argument?

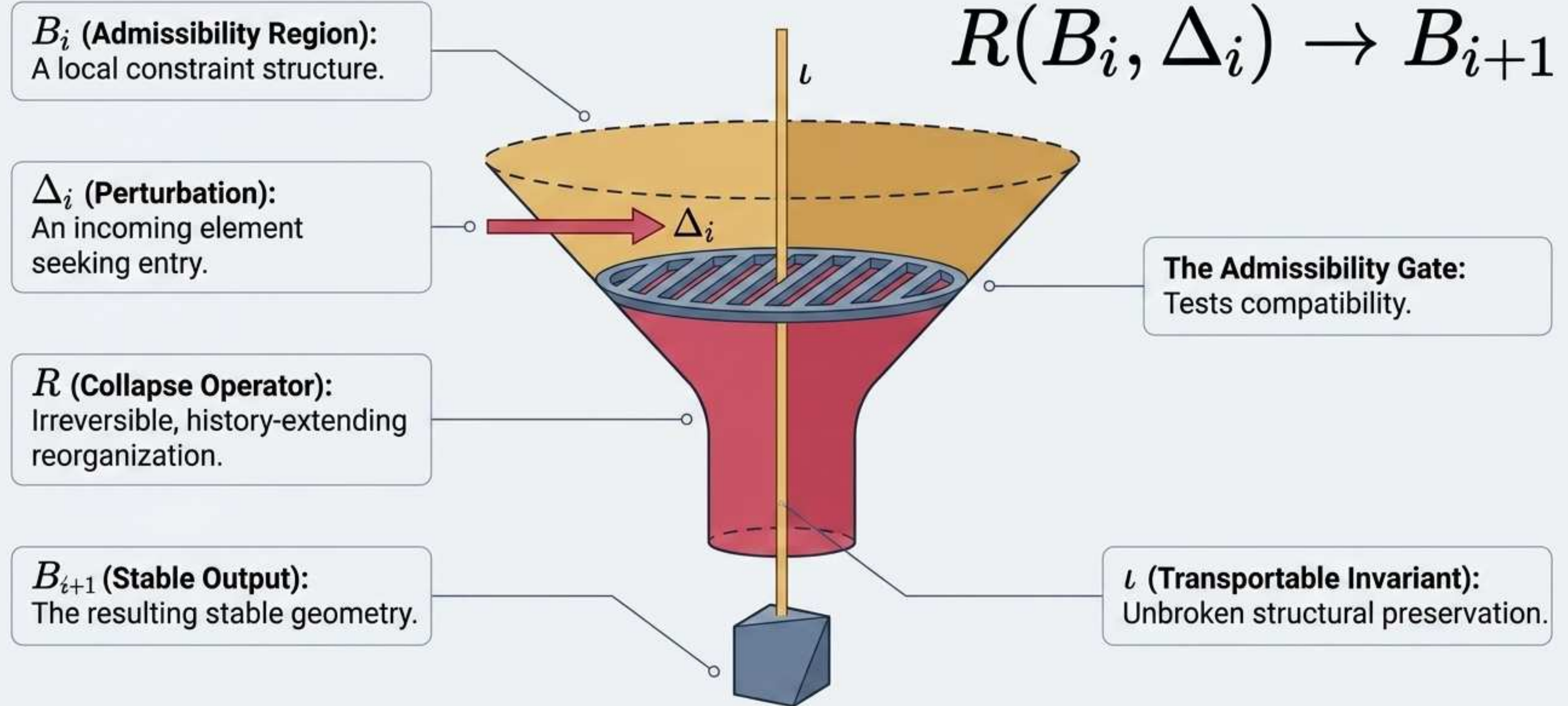


Cognition

What does a mind do when a perception activates a stored conceptual chunk?



The Generalized Reduction Operator



Rigorous Boundaries: What Does Not Qualify

The Admittance Checklist

- ✓ 1. Recursive Restructuring
- ✓ 2. Transportable Invariant (ι)
- ✓ 3. Path-Sensitive History



The Thermostat

Constrains temperature but does not recursively restructure the space of future trajectories. Mere constraint is not admissibility.



Shattered Glass

Physical destruction transforms the state, but no coherent, transportable invariant survives to constrain future evolution.

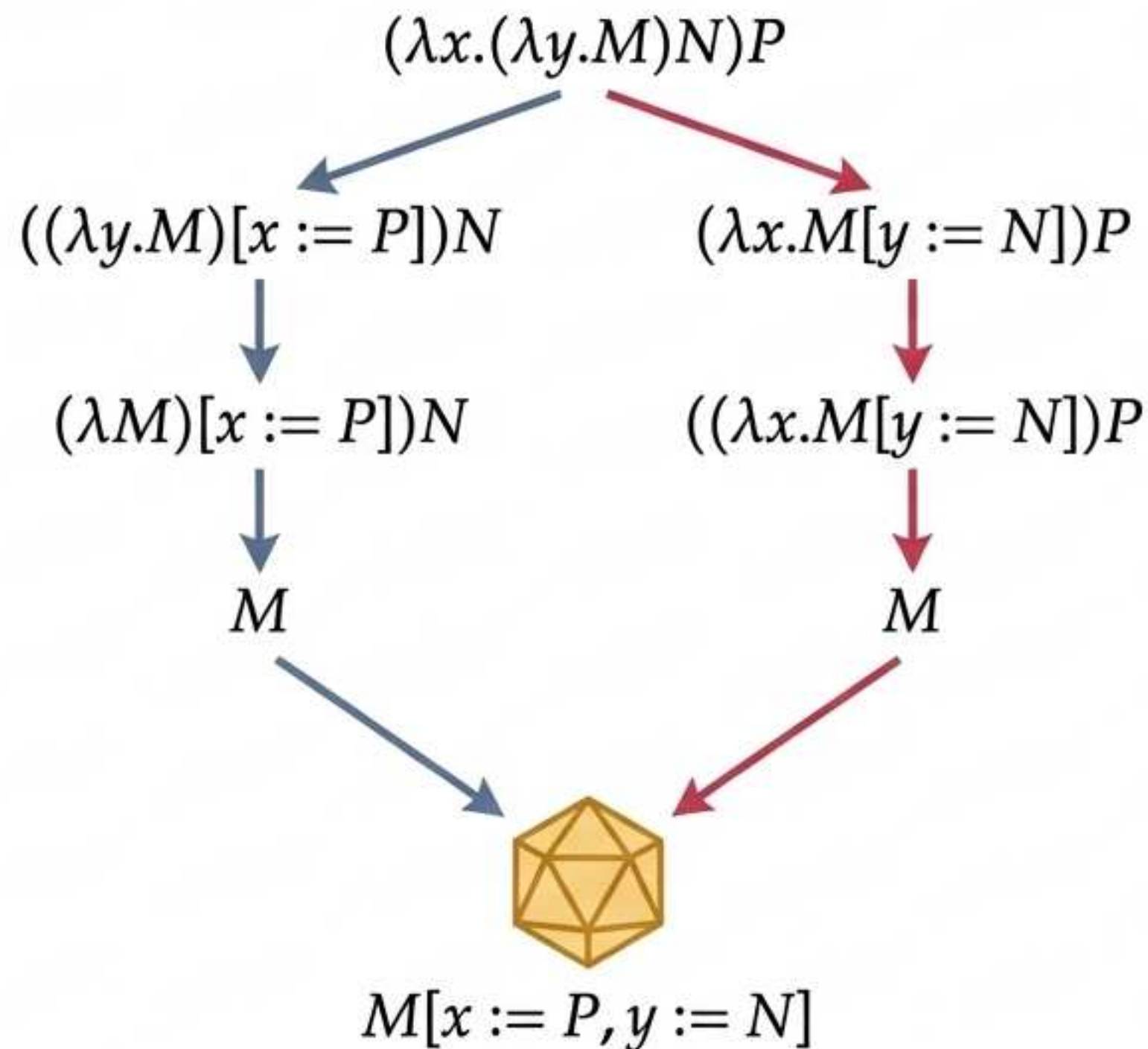


Markov Random Walk

Transitions are local but lack path-sensitive historical dependence.

The Syntactic Foundation: λ -Calculus and Confluence

Epistemic Status: Established (peer-reviewed results)
The results in this section are standard and fully established.
Primary sources: Church (1936), Curry and Feys (1958),
Howard (1980), Girard (1986), and Sørensen–Urzyczyn (2006).

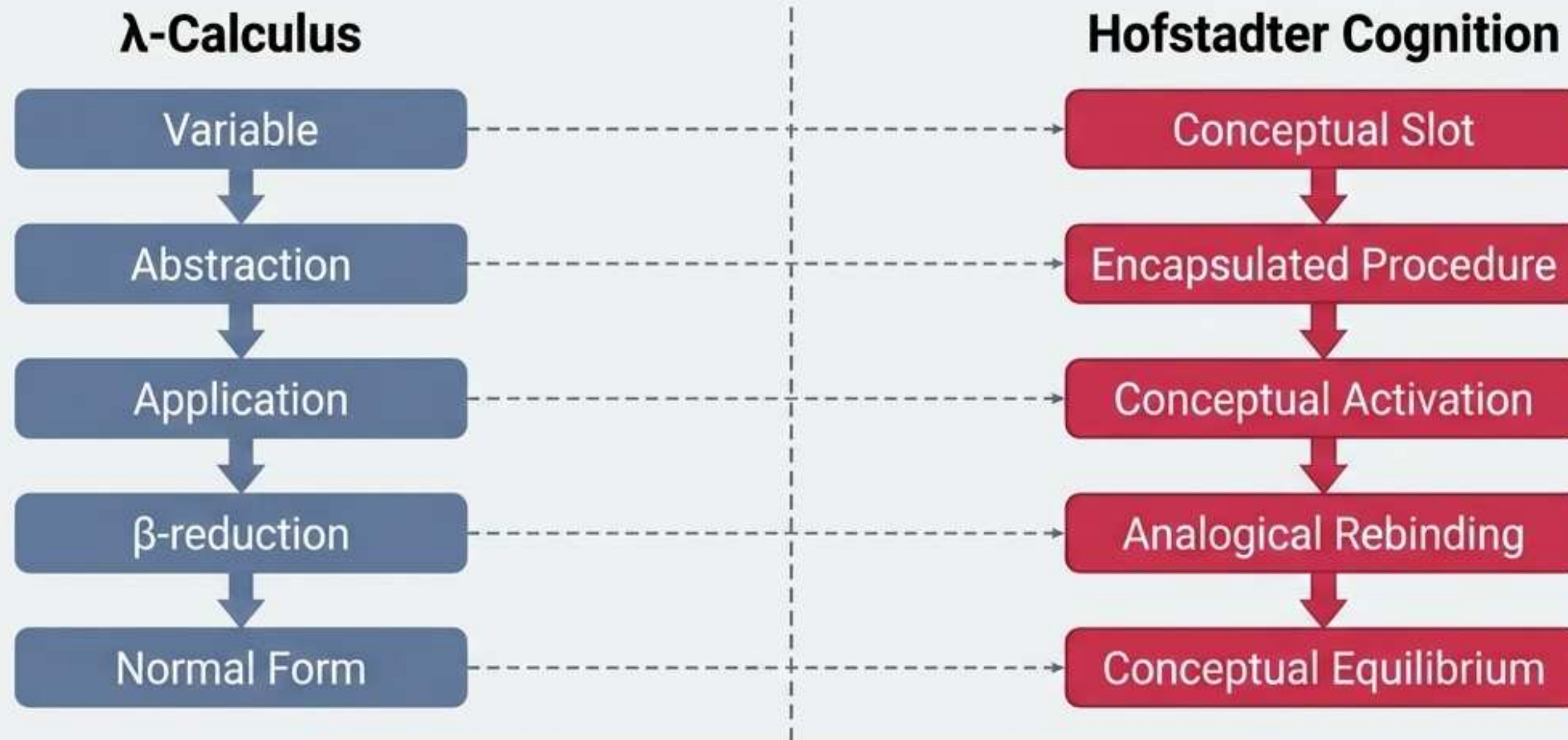


Admissibility Mapping

- 1 **Region:** Scoped abstraction ($\lambda x. M$)
- 2 **Perturbation:** Argument (N)
- 3 **Collapse:** β -reduction ($M[x := N]$)
- 4 **Invariant:** Types (Curry-Howard proofs)

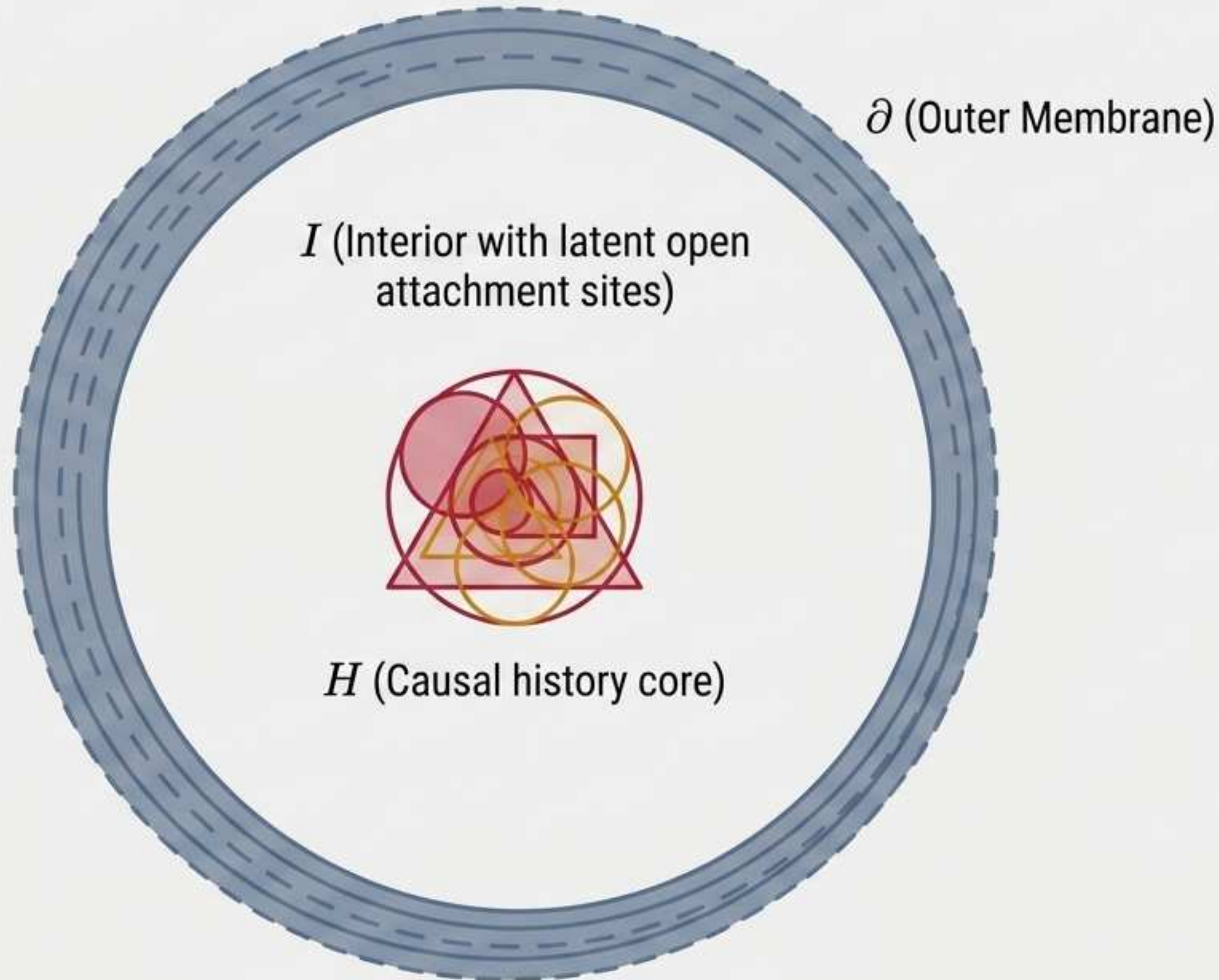
Takeaway: In λ -calculus, incompatible substitution orders are reconciled through normalization to a common normal form. This is the master convergence principle in its purest state.

Cognition as Admissibility-Driven Reduction



Analogy is admissibility-preserving transport.
 β -reduction is its symbolic special case.

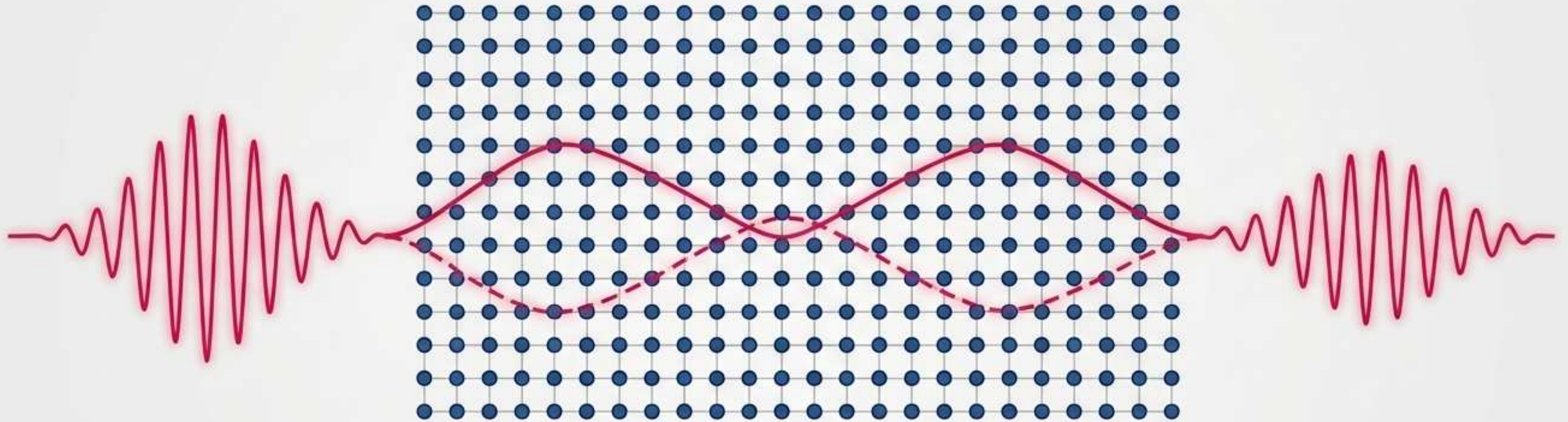
Spherepop: Externalizing Substitution as Topology



Standard λ -calculus hides substitution as a meta-operation. Spherepop treats it as a spatial, topological event where boundaries are first-class objects. Membrane collapse ($b \oplus b' \rightarrow b''$) makes substitution a visible, irreversible history.

Quantum Interference as Admissibility Filtering

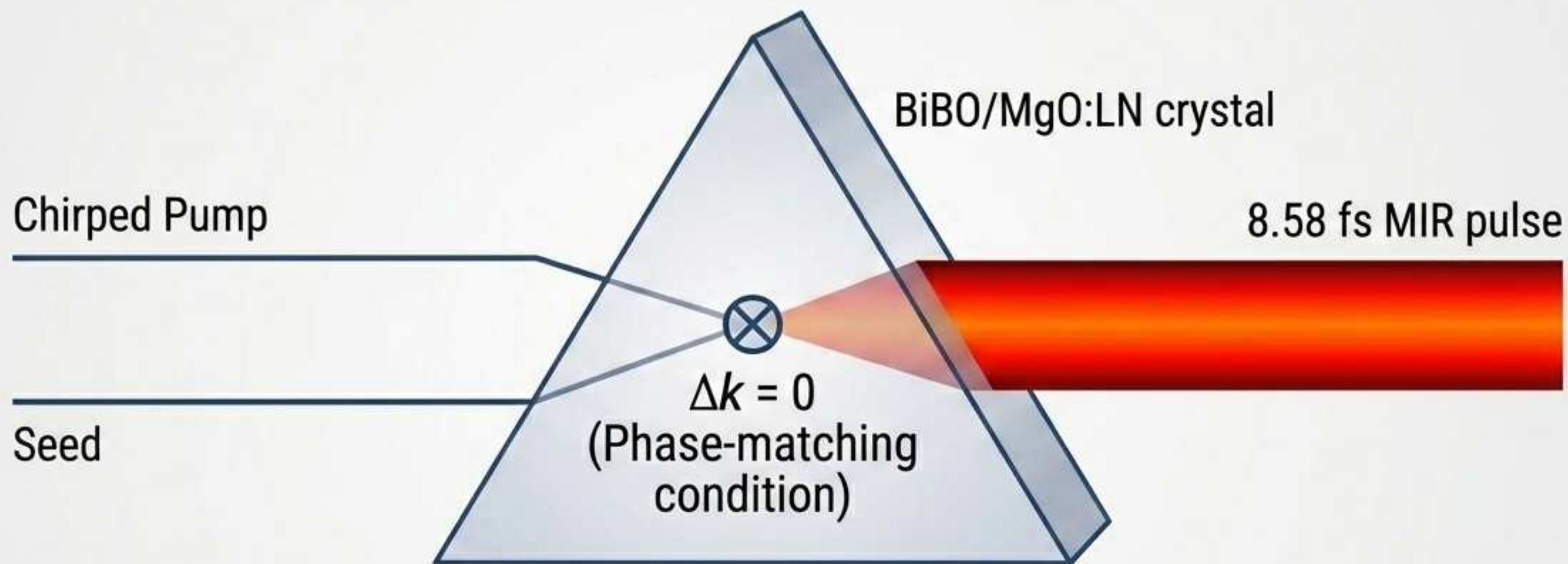
Epistemic Status: Established
(peer-reviewed results)



When a photon briefly becomes an atomic excitation, the weak-value dwell time is determined by an **interference-mediated admissibility gate**. Only trajectories conforming to the measurement protocol survive. Unitarity is the invariant.

Macroscopic Energy Redistribution in Ultrafast Optics

Epistemic Status: Established
(peer-reviewed results)

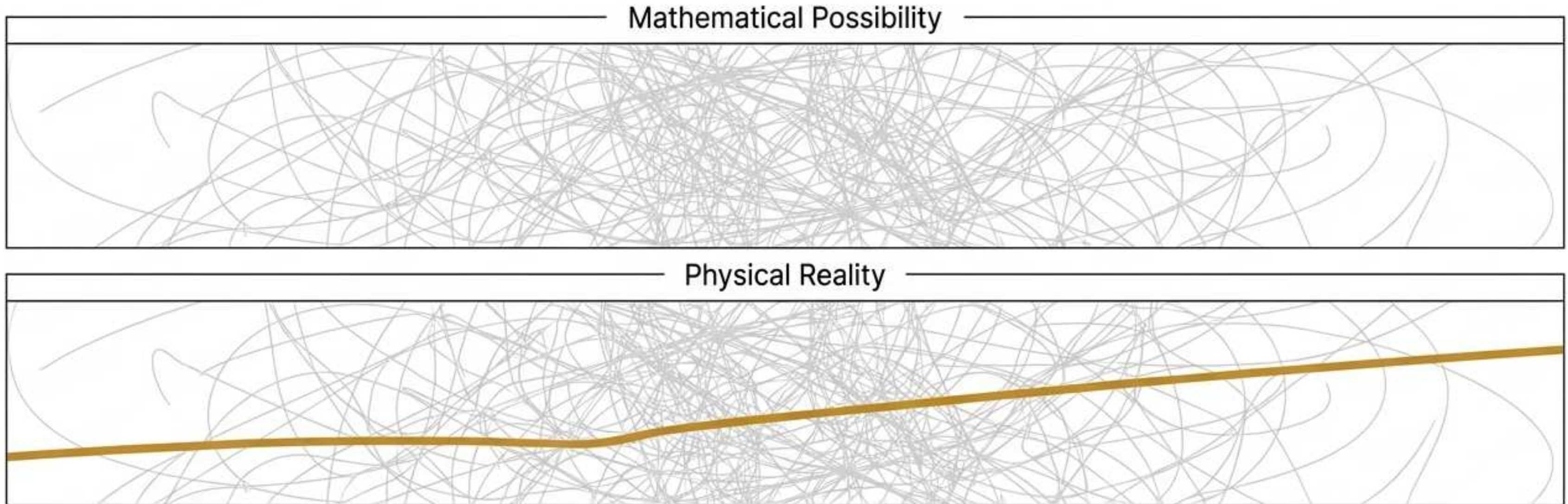


Dual-chirped optical parametric amplification (DC-OPA) requires exact phase-matching. The nonlinear crystal acts as the **admissibility region**. Energy is transferred only when the strict structural constraint ($\Delta k = 0$) is met, yielding CEP-stable compressed output.

Thermodynamic Admissibility: Selecting Reality

Epistemic Status:
Speculative Theoretical Proposal

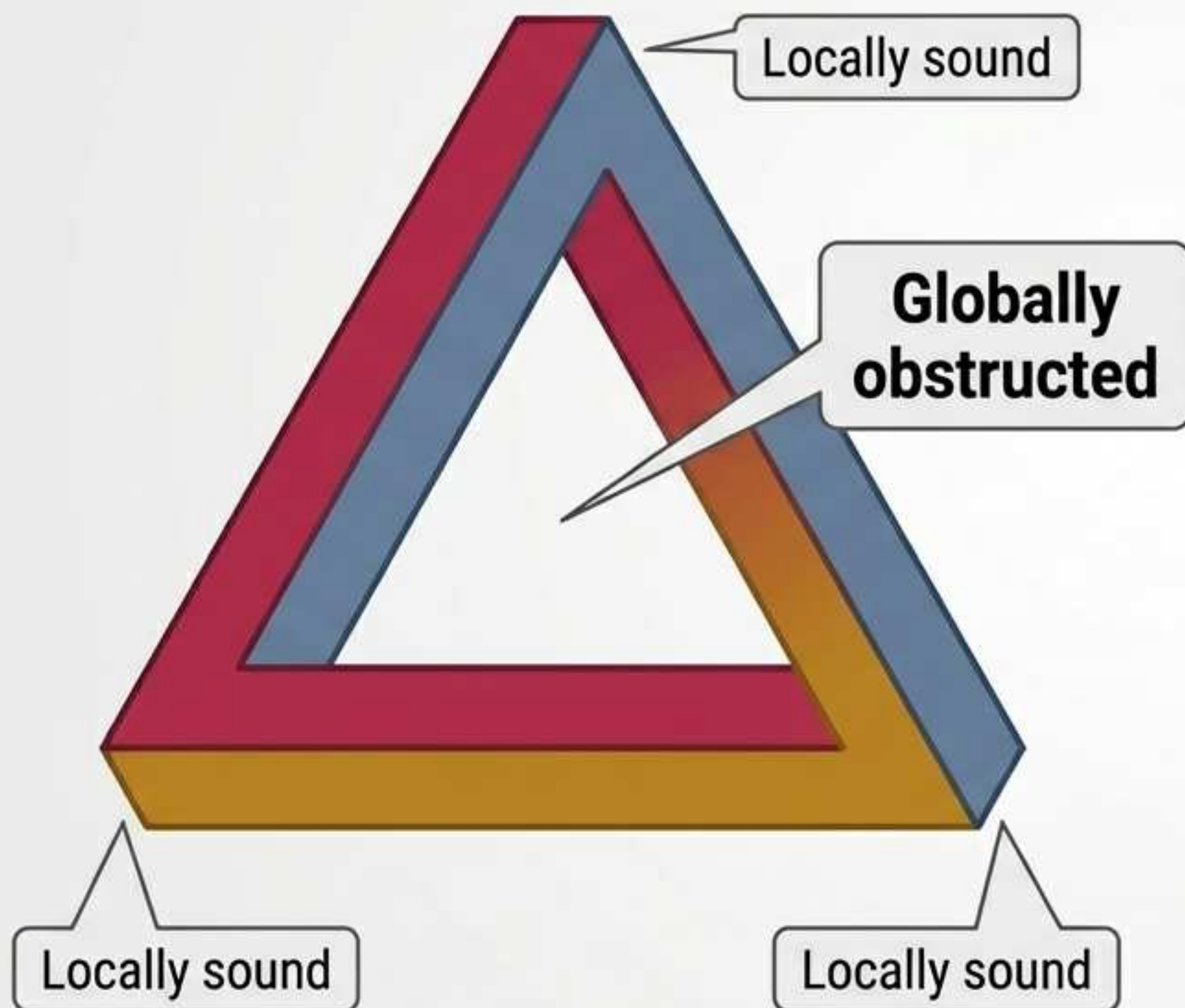
Epistemic Status: Established
(peer-reviewed results)



Dynamics alone are insufficient. In fluid dynamics, maximum entropy production acts as the admissibility selector, pruning mathematically valid but physically incorrect trajectories.

Similarly, in the RSVP framework, entropy bounds constrain field relaxation, preventing supercritical runaway.

The Gluing Obstruction: Why Systems Hallucinate



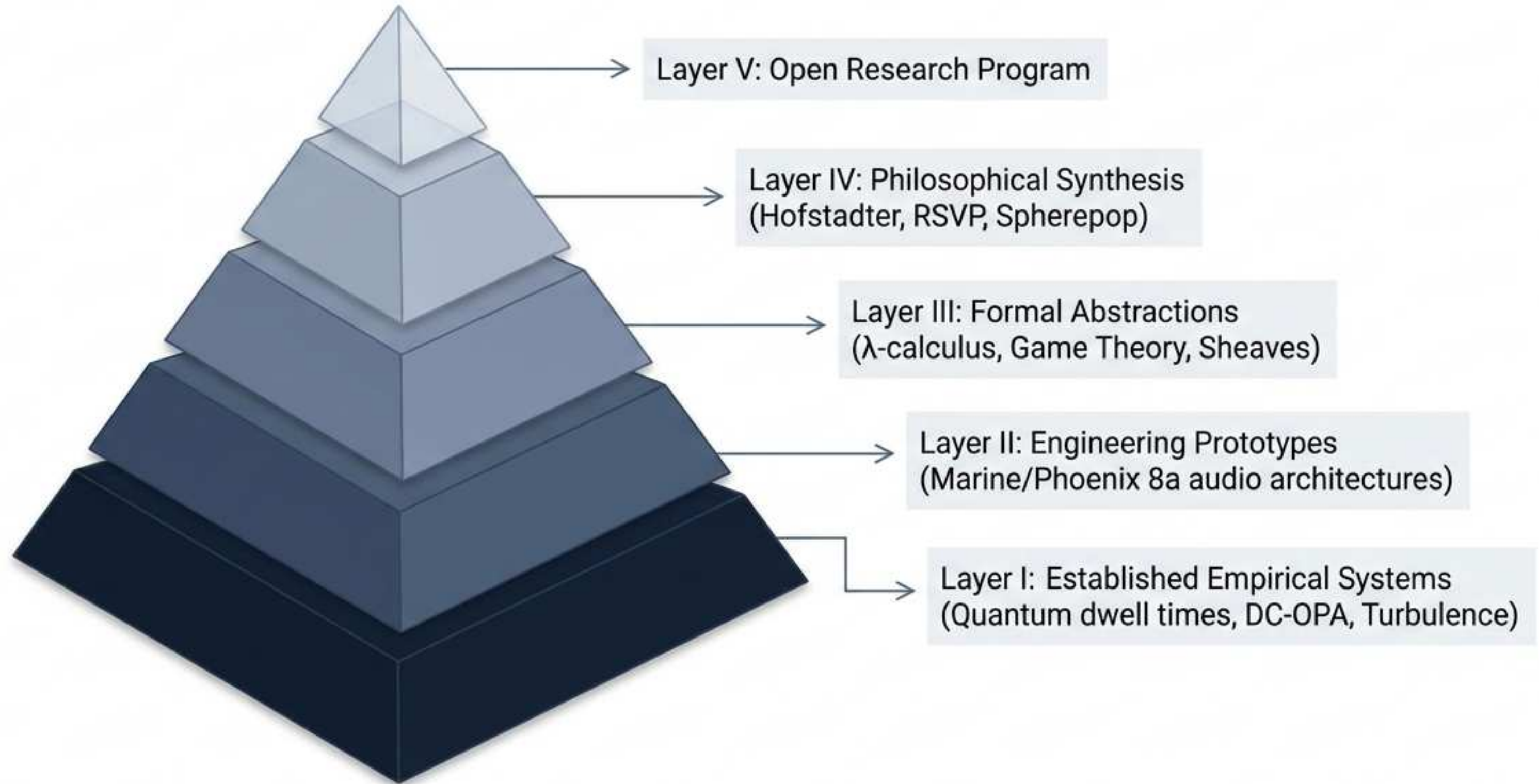
Sheaf Theory

Local patches are mathematically consistent, but no global section exists.

Cognition & LLMs

Individual analogical fragments or token sequences are generated with high local plausibility. However, they fail the cocycle condition—they cannot glue together into a global factual structure that respects world-knowledge constraints.

The Five-Layer Architecture



The Universal Concept Atlas

Epistemic Status: Established
(peer-reviewed results)

Domain	Region	Perturbation	Collapse Operator	Invariant
λ -Calculus	Scoped abstraction	Argument	β -reduction	Types / Proofs
Cognition	Conceptual chunk	Perceptual input	Analogical rebinding	Stable concept
Quantum Optics	Atomic medium	Photon wavepacket	Destructive interference	Unitarity
DC-OPA	Nonlinear crystal	Pump + seed pulses	3-wave mixing	Photon number
RSVP Dynamics	Field region	Entropy flux	Gradient relaxation	Total entropy
Game Theory	Information set	New strategy	Iterated weak dominance	Representation

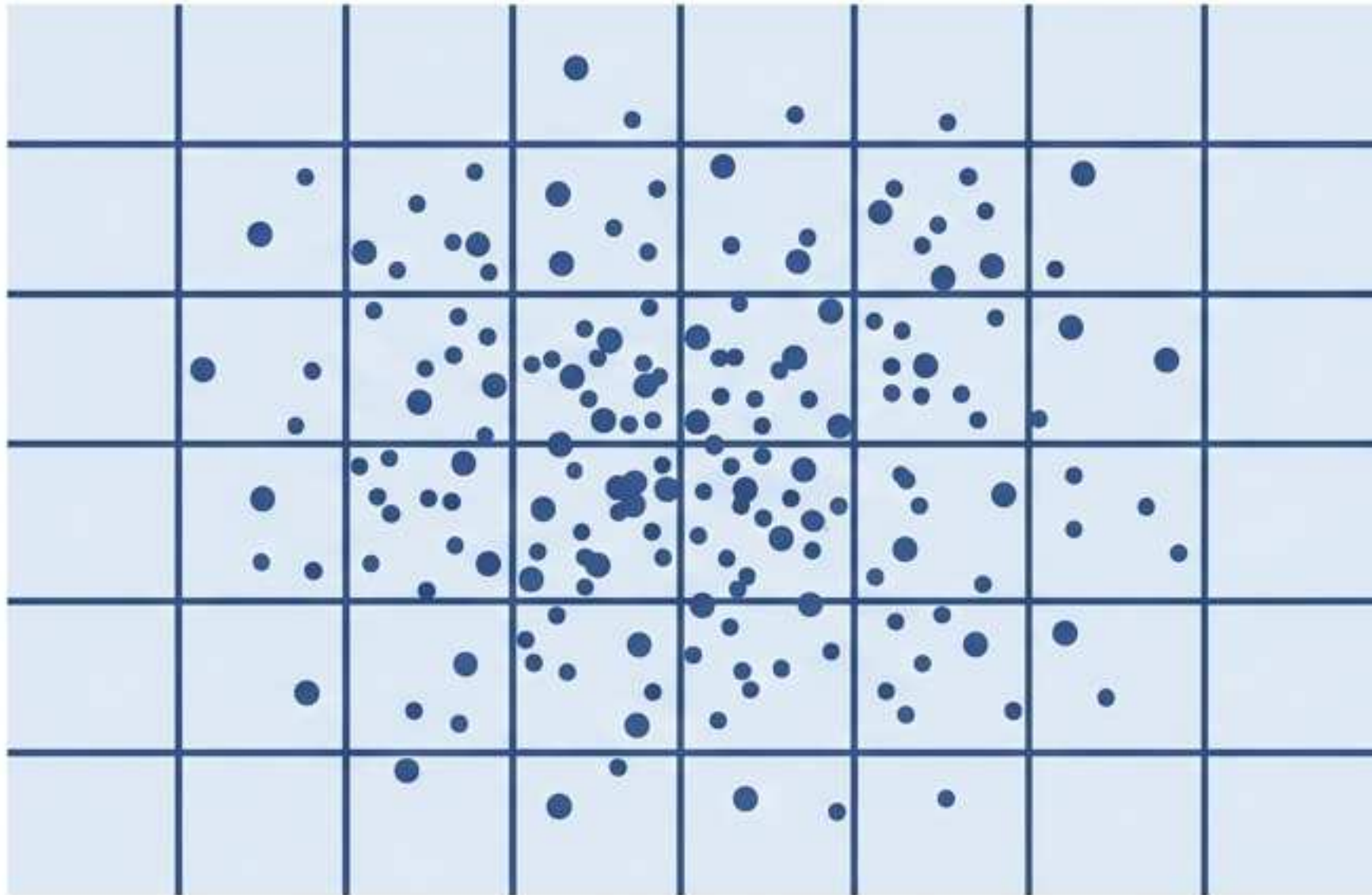
The Philosophical Inversion

Epistemic Status: Speculative Theoretical Proposal
The Modgil-Patil (2026) paper on observer admissibility and Planck invariance is not yet peer-reviewed in its current form and contains acknowledged mathematical gaps (particularly in the covariant formulation of s_h and the phenomenological global-bound formula). The Evangelista (preprint) and Zenteno (Zenodo, 2026) papers are similarly pre-publication. The material in this section is presented as a convergent theoretical program, not established physics. The mapping onto the admissibility schema is the authors' synthesis.

Epistemic Status: Speculative Theoretical Proposal
This section integrates the admissibility program with several Flynn frameworks (RSVP, CLIO, TARTAN, Veresewler) that are under active theoretical development. The connections drawn here are interpretive proposals, not established results.

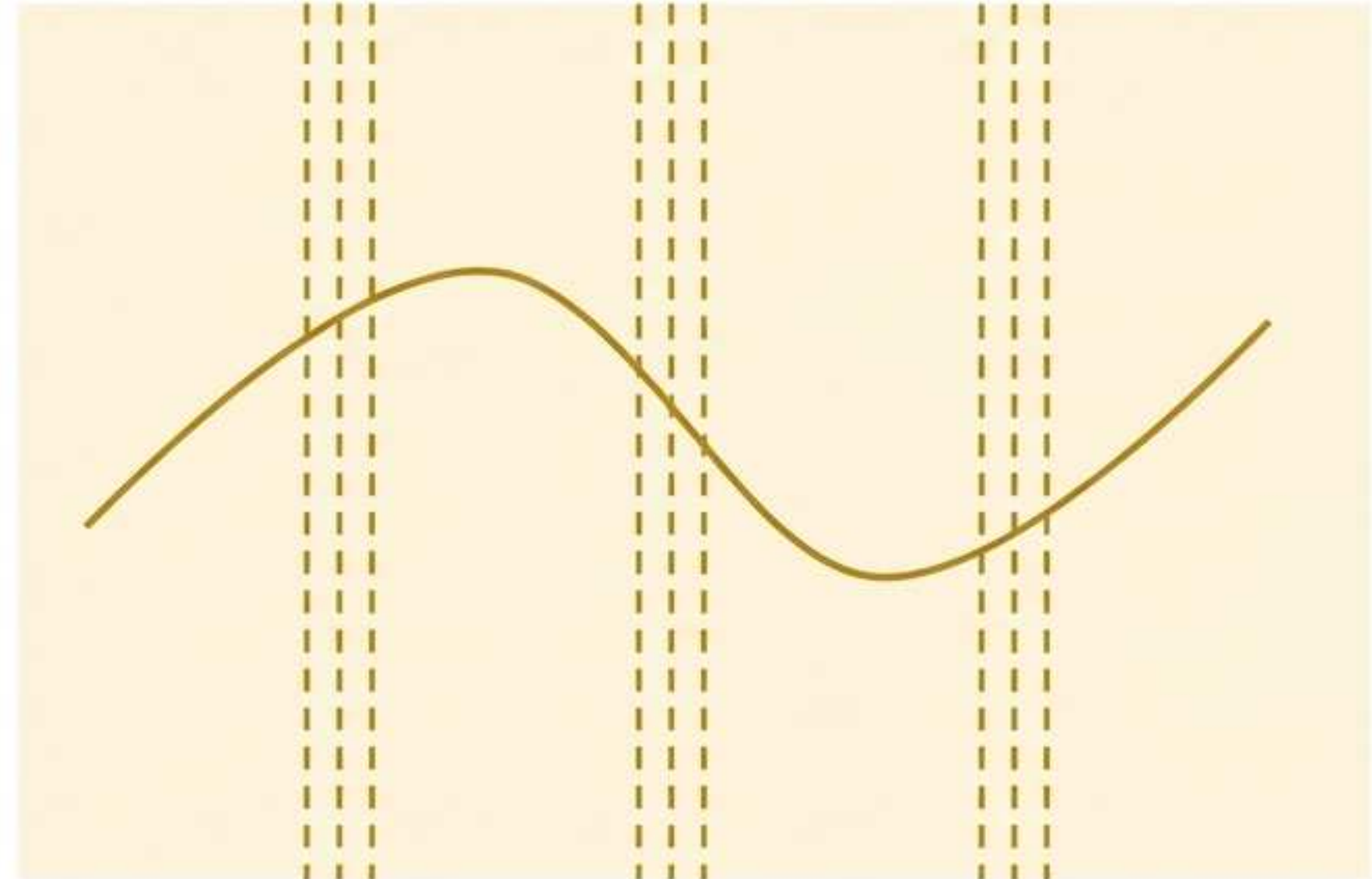
The Classical View

Physics governs what exists.



The Admissibility View

Physics governs what remains coherently observable.



Admissibility is not an additional force. It is a meta-constraint on existence. The universe is a dynamic filter selecting which trajectories can persist without collapsing their own conditions of observation.

A Shared Research Grammar

The Admissibility Principle is not a final physical law or a mathematical theorem. It is a research grammar—a way of asking a universal question:

- What makes a perturbation locally receivable?
- What becomes stable once it has been received?